

Naukri Job Market Analytics Dashboard

Power BI Project Report

Meet Ghadiya | April 2026 | github.com/MeetGhadiya/PowerBI-Dashboard

60,645	8	4	11.14 LPA
Total Job Postings	Job Categories	Dashboard Pages	Avg Salary (Disclosed)

Tools: Power BI Desktop | Power Query (ETL) | DAX | Python (Pandas) | GitHub

1. Project Overview

This project is an end-to-end Power BI analytics dashboard built on 60,645 real job postings scraped from Naukri.com across 8 job categories. The goal was to uncover key trends in the Indian job market — including top hiring companies, in-demand skills, salary benchmarks, and city-wise demand — and present them in an interactive, portfolio-ready dashboard.

Project Objectives

- Analyze 60,000+ job postings across Data Science, Engineering, IT, and more
- Identify top hiring companies, cities, and in-demand skills
- Build a professional 4-page Power BI dashboard for portfolio use
- Demonstrate ETL, DAX, and data modeling skills end-to-end
- Publish to GitHub as a structured, documented portfolio project

2. Dataset Details

Attribute	Details
Source	Naukri.com (scraped dataset)
Total Records	60,645 job postings
Categories	8 (Data Science, Software Engineering, IT Security, etc.)
Key Columns	Title, Company, Salary, Experience, Location, Skills
Skill Columns	Skill_1 to Skill_8 (up to 8 skills per job posting)
Salary Disclosed	5,880 out of 60,645 (9.7% disclosure rate)
Top Cities	Bengaluru, Hyderabad, Pune, Mumbai, Chennai
Data Cleaned Using	Python (Pandas) + Power Query in Power BI

3. Data Pipeline & ETL

The data pipeline followed a three-step process from raw CSV files to a clean, analysis-ready Power BI model.

Step 1 — Data Collection & Merging (Python)

Eight separate CSV files (one per job category) were merged into a single master dataset using Python (Pandas). A Category column was appended to each record to track its source category.

Step 2 — Data Cleaning (Python)

- Renamed messy column headers (e.g., 'dot-gt' → 'Skill_1')
- Filled null salary values with 'Not Disclosed'
- Extracted numeric mid-salary value from text ranges (e.g., '10-20 Lacs PA' → 15.0)
- Parsed minimum experience years from text-based ranges
- Extracted primary city from full location strings

Step 3 — Power Query ETL (Power BI)

- Set correct data types for Rating, Exp_Min_Years, and Salary_Mid_LPA columns
- Unpivoted Skill_1 through Skill_8 into a single normalized Skill column
- Trimmed and lowercased all skill values for consistent aggregation
- Filtered out blank skill rows to ensure clean visuals

4. DAX Measures

Custom DAX measures were created to power KPI cards and calculated insights across the dashboard.

Measure Name	Formula	Purpose
Total Jobs	COUNTROWS(Table)	Count of all job postings
Avg Salary LPA	AVERAGE(Salary_Mid_LPA)	Average salary among disclosed records
High Pay %	DIVIDE(CALCULATE([Total Jobs], Salary > 12), [Total Jobs]) * 100	% of jobs paying above 12 LPA
Salary Disclosed %	DIVIDE(CALCULATE([Total Jobs], Salary <> "Not Disclosed"), [Total Jobs])	Salary transparency rate

5. Dashboard Pages

Page 1: Overview

- **KPI Cards:** Total Jobs (485K), Avg Salary LPA (11.14), High Pay % (3.49%)
- **Bar Chart:** Total Jobs by Category — Engineering Software leads with 330K
- **Bar Chart:** Total Jobs by Title — Application Developer tops with 81K

Page 2: Roles & Companies

- **Bar Chart:** Top 10 Companies — Accenture leads with 180K postings
- **Scatter Plot:** Experience vs Salary by Category
- **Bar Chart:** Top 10 Job Titles by posting count

Page 3: Skills Analysis

- **Treemap:** Top 20 Skills — SQL, Python, and Java dominate
- **Clustered Bar:** Top skills breakdown by job category
- **Insight:** CSS ranks high due to the UX/Design category presence

Page 4: Market Insights

- **Bar Chart:** Top 10 Cities — Bengaluru leads with 142K jobs
- **Bar Chart:** Avg Salary by City — Jamnagar tops at 53 LPA
- **Bar Chart:** Jobs by Experience — 3–5 Years is the most demanded band (53K)
- **Summary Table:** Top companies ranked by total job count

6. Key Insights

Bengaluru is the #1 Job Hub

With 142,000 postings, Bengaluru accounts for nearly 23% of all jobs in the dataset, followed by Hyderabad (70K) and Pune (57K).

Engineering Dominates the Market

The Engineering Software & QA category alone contributes 330,000+ postings — approximately 55% of the entire dataset.

3–5 Years Experience is Most In-Demand

53,000 jobs target candidates with 3–5 years of experience, making it the most competitive experience band in the Indian market.

SQL, Python & Java Lead In-Demand Skills

SQL appears in the highest number of job postings, followed by Python and Java. These three skills collectively cover the majority of tech roles.

Salary Transparency Remains Low

Only 9.7% of jobs (5,880 out of 60,645) disclose salary information. Among those that do, the average is 11.14 LPA.

Accenture is the Dominant Recruiter

Accenture dominates with 180,888 job postings — nearly 30x more than the second-largest recruiter, IBM (6,088 postings).

7. GitHub Project Structure

The project is publicly available on GitHub at: github.com/MeetGhadiya/PowerBI-Dashboard

Repository: PowerBI-Dashboard / Naukri-Job-Market-Analysis

File / Folder	Description
Naukri_Job_Market_Dashboard.pbix	Main Power BI dashboard file (4 pages)
Naukri_Combined_AllCategories.csv	Merged and cleaned dataset (60,645 rows)
Naukri-Job-Market-Analysis.PNG/	Folder containing dashboard screenshot exports (Insights, Overview, Roles, Skills)
README.md	Project documentation and setup instructions
Project_Report.pdf	This report in PDF format

8. Tools & Technologies

Tool / Technology	Usage
Power BI Desktop	Dashboard creation, data modelling, DAX measures
Power Query (M Language)	ETL — data type fixes, skill column unpivoting, filtering
DAX (Data Analysis Expressions)	Custom calculated measures for KPIs and insights
Python (Pandas)	CSV merging, data cleaning, salary & experience parsing
GitHub	Version control and public portfolio hosting

9. Conclusion

This project demonstrates a complete, end-to-end data analytics workflow — from raw data collection and Python-based cleaning, through Power Query ETL and DAX measure creation, to building a professional multi-page Power BI dashboard.

The insights derived from 60,000+ job postings provide actionable intelligence for job seekers, recruiters, and data professionals operating in the Indian job market. Key findings reveal that Bengaluru dominates hiring, Engineering Software is the largest sector, and SQL/Python/Java are the foundational skills demanded across roles.

As a portfolio project, this dashboard showcases proficiency across the full data analyst toolstack and reflects the ability to communicate data-driven insights in a clear, visually compelling format.